

Objectives:

- Write MPI programs

Exercise 0:

Please complete the in class activities we did in the MPI lecture given at <https://github.com/SLIIT-FacultyOfComputing/SE4060-MPI-Part1>

Exercise 1:

- Create a GitHub repo with your student id. If you already have one you can use it.
- Login to your GitHub account
- You have to do the exercises given in the repository and commit and push your solutions into it.

Exercise 2:

Parallelize adding up numbers 1 to 10,000,000 using multiple nodes using MPI

Exercise 3:

Parallelize the computation of Pi using the “Monte Carlo method” (See sample C code in the following link) using MPI

http://www.dartmouth.edu/~rc/classes/soft_dev/C_simple_ex.html

Find the calculation for 10,000,000 times.

Exercise 4:

For both Exercise 1 and Exercise 2

1. Plot a graph that shows Time vs Number of Processors
2. Plot a graph that shows Speedup

Exercise 5:

1. Write a simple program where the source and the destination doesn't match. You can take either message1.cc or message2.cc and modify the program. What happens in this situation?
2. Rewrite message2.cc using Buffered Send (BSend) ensure that you do not replace the variables that you are using to send the data.

Exercise 6:

Rewrite Exercise 3 by changing the receiving Tag to MPI_ANY_SOURCE, redo Exercise 3 and compare with the original results.

Exercise 7:

Rewrite Exercise 6 by using Buffered Send (BSend).